Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/29/2006
PI Name:	Triplett, Jason Ph.D.		
Project Title:	Investigating the role of alpha-actinin in osteoblast mechanotransduction		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Bone loss associated with exposure to microgravity occurs due to a lack of mechanical loading. The cellular mechanisms by which bone cells sense and respond to mechanical stimuli remain unclear. Our laboratory has shown that focal adhesions play an important role in mechanotransduction in osteoblasts. I am proposing to study the role of alpha-actinin in osteoblast mechanotransduction. Based on alpha-actinin's ability to interact with multiple focal adhesion proteins, it could potentially play an important role in focal adhesion organization and mechanotransduction. I will use a dominant-negative alpha-actinin to assess alpha-actinin's role in focal adhesion organization and mechanotransduction.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Understanding how bone cells sense and respond to mechanical stimuli can help to identify molecular targets to offset the loss associated with long-term space flight. Any manned mission to Mars, for instance, will need to address the issue of bone loss associated with a micro-gravity environment. This may require a combination of molecular medicine and the physical generation of "artificial gravity."
Task Progress:	Within the past year, our lab has made significant progress towards the completion of the aims of our research proposal. First, we have published a paper describing a new technique for improving adenoviral infection of cultured cells (Triplett, et. al. Biotechniques, Dec 2005). Previous studies have shown that the use of DNA-damaging agents, such as topoisomerase inhibitors, increased the expression of adeno-associated viral transgenes and effectively treated cancers in vivo. In our study, we explored the possibility of using low doses of the topoisomerase inhibitor etoposide to increase adenoviral transgene expression in cell culture models. Indeed, we found that use of as little as 100 nM etoposide during adenoviral incubation increased expression ~3 fold in multiple cell types. Use of etoposide also appeared to increase the number of cells expressing transgenes. Furthermore, no averse effects on cell viability or cell function were observed. We believe this technique represents a simple, cost-effective means to increase adenoviral infection efficiency, especially in resistant cell types, such as cultured osteoblasts. Additionally, we have published a paper in which we explored the effects of expression of a truncated version of a-actinin on focal adhesion organization and cell survival (Triplett and Pavalko. Am J Physio – Cell Physio, in press). This construct, termed ROD-GFP, effectively disrupts focal adhesion organization, as evidenced by a reduction in focal adhesion size. ROD-GFP expression also results in inhibited focal adhesion signaling capabilities, evidenced by decreased tyrosine phosphorylation and focal adhesion kinase (FAK) phosphorylation. Further more, we demonstrated that expression of ROD-GFP renders cells more susceptible to apoptosis induced by tumor necrosis factor alpha (TNFa). We showed that this sensitization was due to decreased expression of anti-apoptotic proteins, such as Bcl-2 and Bcl-XL, and decreased survival signaling through the TNFa receptor pathway. These results are of particular interest, because regulation of apoptosis at sites of bone formation have been shown to be important in the ability of bone to respond to mechanical stimuli properly. In conclusion, our studies over the past year have yielded interesting and important results concerning the regulation of osteoblast survival by a-actinin and focal adhesions. We believe that our efforts have provided the field with novel tools to aid in researching the role of cell adhesion in osteoblast physiology, which could lead to the development of countermeasures to the loss of bone associated with spaceflight.
Bibliography Type:	Description: (Last Updated: 01/08/2020)
Articles in Peer-reviewed Journals	Triplett JW, Herring BP, Pavalko FM "Adenoviral transgene expression enhanced by co-treatment with etoposide in cultured cells" Biotechniques. 2005 Dec;39(6):826,828,830, passim. PMID: 16382899 , Dec-2005